FUGACITY 2026

Instructions

- **Choose any one** of the given problem statements and develop your solution around it.
- The final pitch deck must be submitted in **PDF format through the registration portal exactly 24 hours before the event**. No late submissions or on-the-spot file changes will be accepted.
- The pitch must not exceed the **5-minute time limit, followed by a QnA session**. Exceeding the limit will result in point deductions.
- The solution must be the **original work of the participant/team**. Plagiarism, fabricated data, or copying existing solutions will result in disqualification.
- **All literature values and references** used in the presentation must be cited on the last slide.
- **AI usage is permitted only for images**, with proper disclosure. Use of AI-generated content for any other part of the presentation will result in disqualification.
- The decision of the **faculty jury and Fugacity 2026 Organizing Committee** will be final and binding.

PROBLEM STATEMENTS

1. Green & Sustainable Technology

Problem Statement

Chemical manufacturing is often associated with high energy consumption, greenhouse gas emissions, and significant environmental impact. At the same time, industries need to maintain production levels, product quality, and profitability.

How can you redesign, optimize, or improve a chemical manufacturing process to reduce its carbon footprint, energy consumption, and environmental impact while maintaining productivity, product quality, and economic viability?

Participants are encouraged to consider practical solutions that can be implemented at an industrial scale, along with the technical and economic feasibility of their proposed approach.

2. Next-Generation Battery Materials

Problem Statement

The growing adoption of electric vehicles, renewable energy, and portable electronics is increasing the demand for high-performance energy storage systems. However, current battery technologies face challenges related to cost, safety, resource availability, environmental impact, energy density, and lifecycle performance.

How can we develop or identify safer, more sustainable, high-performance, and cost-effective battery materials that can address these challenges and meet the growing demand for energy storage?

Participants may explore new materials, alternative raw materials, improved material compositions, or approaches that can make battery technologies more sustainable and commercially viable.

3. Circular Economy & Waste-to-Wealth

Problem Statement

Large quantities of industrial waste are generated across manufacturing sectors every day. Many of these waste streams contain materials, chemicals, or energy that can potentially be recovered and put to productive use. Treating such streams only as waste can lead to resource loss, disposal costs, and environmental concerns.

How can industrial waste streams be converted into valuable products, materials, chemicals, or energy, creating an economically viable solution while reducing waste and environmental impact?

Participants are encouraged to identify promising waste streams and develop practical pathways for their conversion, recovery, or reuse, while considering technical feasibility, scalability, economics, and environmental benefits.